

Fig. 34: Current versus frequency plot in Android app

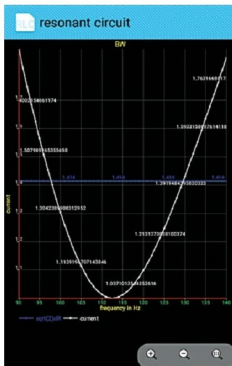


Fig. 36: Bandwidth

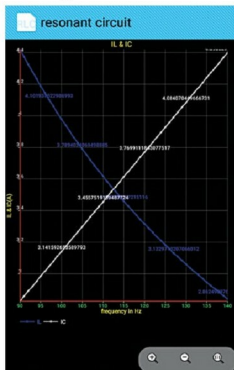


Fig. 35: IL/IC versus frequency

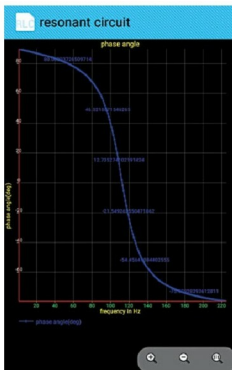


Fig. 37: Phase angle

susceptance graph in Fig. 29.

**Capacitive susceptance (BC).** Capacitive susceptance is denoted by BC and is positive in value. Capacitive susceptance is directly proportional to frequency. In Fig. 30, capacitive susceptance graph indicates BC (linear straight line) as shown in the susceptance graph in Fig. 29.

**Susceptance plot (B).** Susceptance is the reciprocal of reactance

X. At resonance, BL and BC are equal and opposite, so susceptance B becomes zero. In Fig. 31, the white line indicates capacitive susceptance BC, blue line indicates inductive susceptance BL and green line indicates susceptance B.

**Note.** For susceptance plot (Fig. 31), input was modified (Fig. 32) to view the graph clearly.

**Parallel circuit current.** In Fig.

## EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at [source.efymag.com](http://source.efymag.com)

33, at resonance frequency, current is equal to current across the resistor. In Fig. 34, at resonance, current ( $I = 1A$ ) is equal to the resistor current ( $IR = 1A$ ) and this graph is simulated from 90Hz to 140Hz.

**Capacitive and inductive current (IL and IC).** In Fig. 35, at resonance, both IL and IC are equal and opposite in phase, so these cancel each other out and this graph is simulated from 90Hz to 140Hz.

**Parallel circuit bandwidth.** Below resonance frequency, frequency at which the current is  $\sqrt{2}$  times of its minimum value is called lower cut-off frequency. Above resonance frequency, frequency at which the current is  $\sqrt{2}$  times of its minimum value is called higher cut-off frequency. These two cut-off frequencies are called half-power frequencies.

**Bandwidth = higher cut-off frequency - lower cut-off frequency (BW = fH - fL)**

Fig. 36 shows the bandwidth graph simulated from 90Hz to 140Hz. Here, at resonance condition, current  $I = 1A$ , which is equal to resistor current  $IR = 1A$  and lower cut-off frequency  $fL = 97.74Hz$  and higher cut-off frequency  $fH = 129.57Hz$ .

**Bandwidth = higher cut-off frequency - lower cut-off frequency (BW = 129.57Hz - 97.74Hz = 31.8Hz).**

**Phase angle.** At resonance frequency, phase angle will be at  $0^\circ$ . As shown in Fig. 37, at resonance frequency  $f_r = 129.57Hz$ , phase angle is at  $0^\circ$ .

**Note.** The application supports Android operating system starting from Froyo to Lollipop. ●

*Concluded*



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